

01EST102052002 B

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Pages: 2

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester B.Tech Degree Examination July 2021 (2019 scheme)

Course Code: EST102
Course Name: PROGRAMMING IN C
(AN Session)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all Questions. Each question carries 3 Marks

Marks

- | | | |
|----|---|-----|
| 1 | Differentiate between system software and application software. | (3) |
| 2 | Write an algorithm to find the largest of three numbers | (3) |
| 3 | What is the difference between assignment and equality operators? | (3) |
| 4 | What is a static variable? When should it be used? | (3) |
| 5 | Write a C program to find length of a string without using string handling functions. | (3) |
| 6 | What is an array? Illustrate using an example, how a single dimensional array is initialised. | (3) |
| 7 | Differentiate between structure and union using an example. | (3) |
| 8 | Illustrate the purpose of return statement using an example. | (3) |
| 9 | Differentiate between char name[] and char *name in C. | (3) |
| 10 | Explain the use of fseek() function. | (3) |

PART B

Answer any one Question from each module. Each question carries 14 Marks

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|----|--|-----|
| 11 | a) Draw a flowchart to find the factorial of a number. | (6) |
| | b) With the help of a neat diagram explain the functional units of a computer. | (8) |

OR

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|----|---|-----|
| 12 | a) List five important registers in CPU. Also state the purpose of each register. | (6) |
| | b) Write algorithm and draw flowchart to perform swapping of two numbers. | (8) |
| 13 | a) Explain arithmetic, logical and bitwise operators with examples. | (7) |
| | b) Write a C Program to check if a given number is a strong number or not. A strong number is a number in which the <i>sum of the factorial of the digits is equal to the number itself</i> .
Eg:- $145 = 1! + 4! + 5! = 1 + 24 + 120 = 145$ | (7) |

OR

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|----|--|-----|
| 14 | a) Write C program to convert the given decimal number into binary number. | (7) |
|----|--|-----|

01EST102052002 B

- b) What do you mean by Formatted Input? Explain in detail the prototype of 'scanf()' function in C including its argument list and return type. (7)
- 15 a) Explain any 4 string handling functions in C programming. (7)
- b) Write a C program to perform linear search on an array of numbers (7)

OR

- 16 a) Write a C program to find second largest element in an array. (7)
- b) Write a C program to check whether a string is palindrome or not without using string handling functions. (7)
- 17 a) What are different storage classes in C? Give examples for each. (7)
- b) Write a C program to: (7)
- (i) Create a structure with fields: Name, Address, Date of birth.
- (ii) Read the above details for five students from user and display the details

OR

- 18 a) What is recursion? Write a C program to display Fibonacci series using recursive function. (7)
- b) Write a C program to sort N numbers using functions. (7)
- 19 a) Explain any 5 file handling functions in C. (7)
- b) Write a C program to reverse a string using pointers. (7)

OR

- 20 a) Differentiate between array of pointers and pointer to an array. (7)
- b) Write a C program to count number of lines in a text file. (7)
